May 2026 – Aug 2026

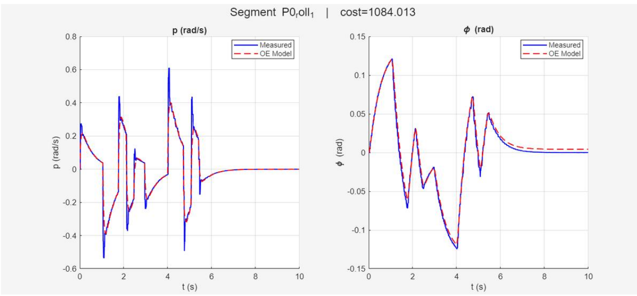
Anduril Industries AI Grand Prix

State Estimation | GNC | Computer Vision | Sensor Fusion | System Identification

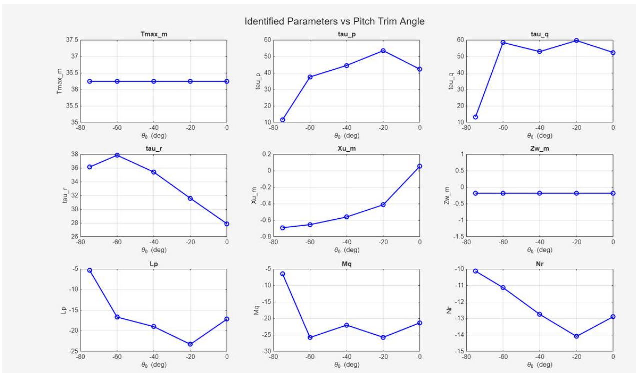
What — A fully autonomous drone pilot stack built for the Anduril Industries AI Grand Prix competition, designed to navigate a racing course from takeoff to landing without any human input.

How — Developed MAVLink communication handlers for telemetry and command, a computer vision pipeline for gate detection, state estimation, and a cascade controller architecture managing thrust and attitude. Conducted a formal system identification study of the drone's flight dynamics to parameterize and tune the control design.

Results — Advanced to the final round of virtual qualifiers, but did not make it to physical competition. The system achieves stable autonomous flight with real-time gate detection and course navigation.



System Identification Diagnostic Plot



Multi-Regime Parameter Visualization

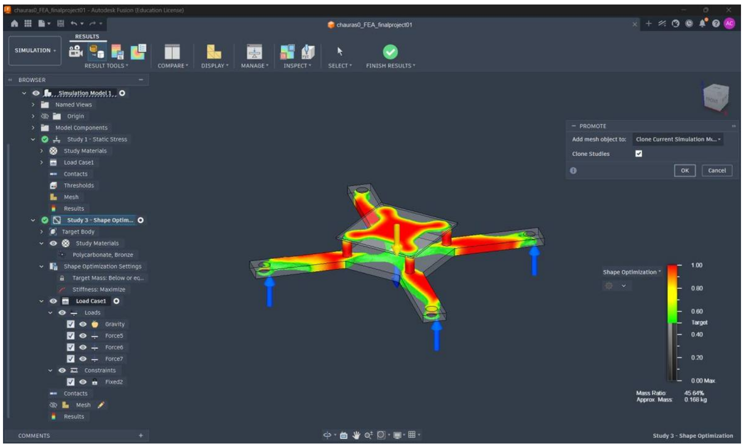
GitHub ↗ https://github.com/ayushc121/AndurilGP

Video ↗ https://www.youtube.com/watch?v=HfdJw55CZkQ

Jun 2026

Drone Frame FEA Study

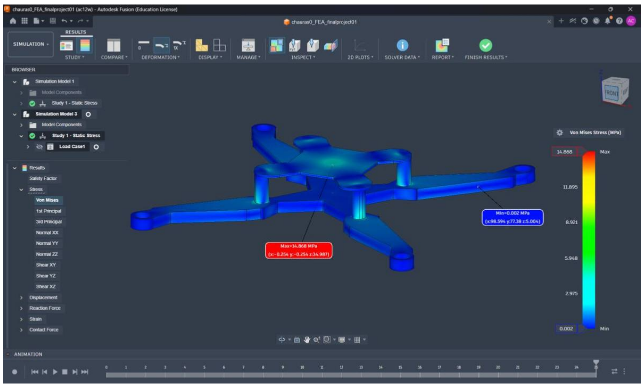
FEA | Shape Optimization | Structural Analysis | Simulation



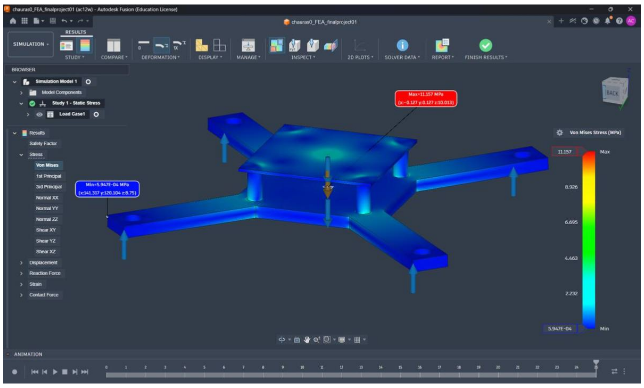
What — A weight minimization study on a multirotor drone airframe, targeting maximum mass reduction while preserving structural integrity and safety margins under realistic loading conditions.

How — Applied iterative shape optimization to progressively refine frame geometry, evaluating stress distributions and safety factors at each design iteration. Validated the final design through dynamic event simulation, modeling high-speed collision scenarios to confirm structural integrity under transient impact loads.

Results — Reduced frame weight by over 50%, arriving at a final polycarbonate frame of 0.17 kg capable of safely supporting payloads exceeding 2.5 kg with maintained factors of safety. Dynamic collision simulation confirmed structural integrity throughout the impact event.



Stress Analysis of Optimized Frame



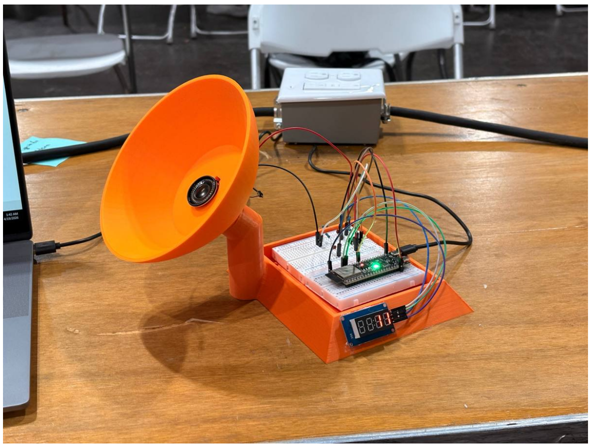
Stress Analysis of Naive Frame

Video ↗ https://youtube.com/watch?v=gPhkBMU2XXw

Apr 2026

Acoustic Drone Detection System

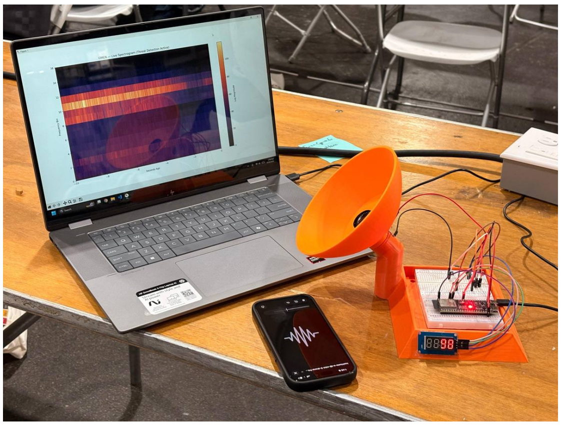
Embedded Systems | C++ | Signal Processing | Machine Learning | PyTorch | Prototyping



What — A low-cost, fully standalone acoustic early-warning sensor for nearby drone detection, designed to operate entirely offline with no cloud or internet dependency.

How — Repurposed an 8Ω speaker as a directional microphone through a custom breadboarded circuit. Programmed an ESP32 microcontroller in C++ to perform real-time FFT analysis on the incoming audio stream, driving hardware alerts with zero latency. Trained a PyTorch neural network on drone acoustic signatures, then embedded the quantized model weights directly onto the microcontroller for on-device inference.

Results — Achieved 80% drone detection accuracy at 15 yards. The complete system (from circuit design through trained model deployment) was built in 36 hours.



Successful Detection of Drone Audio

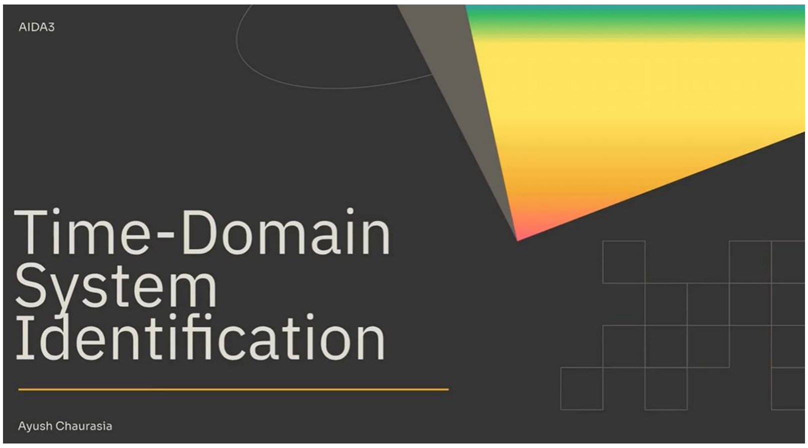
GitHub ↗ https://github.com/ayushc121/starkHacks

Video ↗ https://www.youtube.com/watch?v=E8IrfMaYsBM

Jan 2026 – May 2026

3DoF Plane System Identification Study

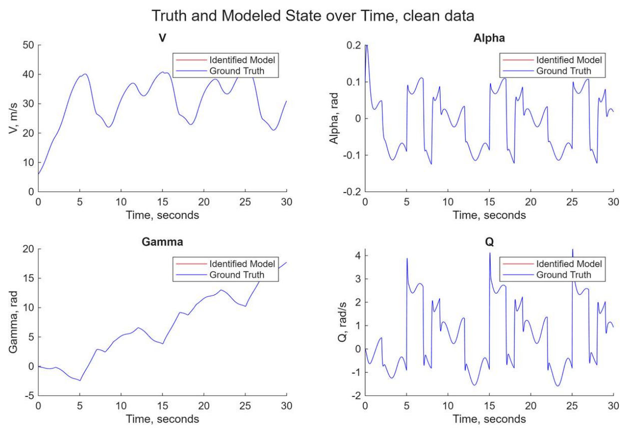
Python | MATLAB | System Identification | Research



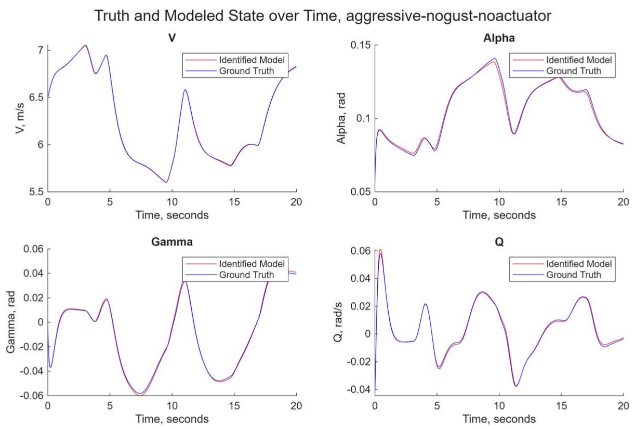
What — A research study evaluating system identification methods for 3-DoF aerodynamic models under noisy and data-limited conditions representative of real flight test environments.

How — Reviewed existing literature and implemented NASA's SIDPAC system identification framework. Built a custom Python-based 3-DoF flight dynamics engine to generate controlled, high-fidelity datasets, then used it to simulate eight targeted edge-case scenarios under external disturbances — enabling systematic evaluation of estimator performance across degraded data conditions.

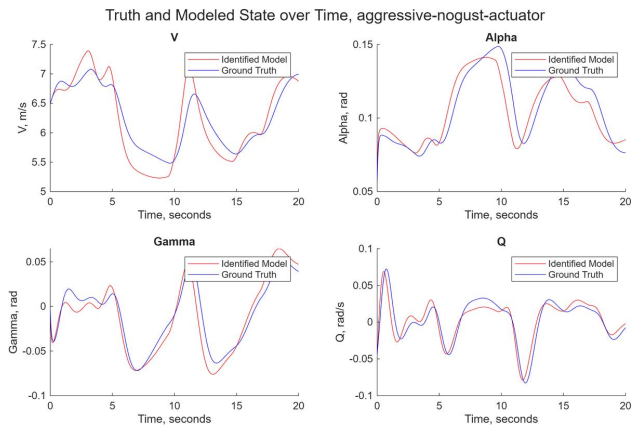
Results — Achieved sub-3% parameter estimation error in noisy 3-DoF aerodynamic simulations with external disturbances using the SIDPAC approach. The custom simulation engine improved testing reliability and provided a repeatable benchmark environment for comparing identification methods.



Perfect Identification Under Ideal Conditions



Good Accuracy Under Realistic Conditions



Moderate Accuracy Under Bad Conditions

GitHub ↗ https://github.com/ayushc121/3dofSysID

Video ↗ https://www.youtube.com/watch?v=ov8d1u_JZXE

Sep 2025 – May 2026

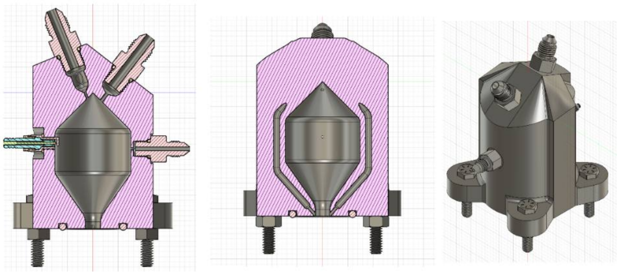
Torch Igniter

Propulsion | Structural Analysis | Manufacturing

What — A torch igniter redesign for a university rocketry club propulsion test system, targeting a significant increase in exhaust temperature without compromising structural safety margins.

How — Led the integration of an oxygen-based afterburner into the igniter assembly, performing structural analysis throughout the design process to maintain sufficient factors of safety. Coordinated with team members and club leadership to plan and execute manufacturing and hotfire testing.

Results — Increased predicted exhaust temperature by 80% over the baseline design. The redesigned igniter passed hotfire testing with structural integrity and safety margins maintained throughout.



CAD Model Views